What the anchor follows at the example answers

> In ex-2.2.22 the alignment with the anchored direction at the example answers rose with the posterior on `difference`, but at the same posterior it sat higher at earlier answers. That difference goes away once each answer is compared with the evidence of its own example. The alignment at an example answer follows how well that one example fits `difference`, and hardly depends on the examples before it.

The backlog item asks whether the anchor holds the op the model has inferred, which would make it grade with the posterior, rather than the binary label it was trained on. Ex-2.2.22 (E3) took a first look and found a rise with the posterior at the example answers, plus an unexplained dependence on the answer index. This page re-analyzes the same stored measurements, with no new runs.

Observations

- Each example on its own (E1): plotted against the posterior given that example alone, the lines for the three answer indices lie on top of each other. On every anchored condition, that posterior accounts for at least 89% of the variance in α (r^2), against at most 48% for the posterior given the examples so far.
- What the earlier examples add (E2): fitted together with the evidence before that example, the earlier evidence gets a weight of at most 0.05, against at least 1.2 for the example itself. It shows most on answers that do not fit `difference`.
- The answer color alone (E3): the answer color, without its operands, predicts at most 9% of the variance, so α depends on how the answer relates to its operands.
- Predicting the next answer (E4): at the query the model does combine the examples. With three examples, the control and `hinge` score 0.88 where the ideal predictor scores 0.95, more than one example alone could support. With one or two examples it falls well short of the ideal predictor, even on `k-mixed`, which was trained on those counts.

Scope

This is a re-analysis of the stored by-count pass of ex-2.2.22 on its three-example held-out set, with no preregistration and no gate. It covers the `hinge` condition (six runs: three paired with ex-2.2.21 and three replicates), the whole-line condition (three runs), `k-mixed` (three runs, one left out as below), and the control (five runs). Only the 2,000 held-out `difference` contexts are used, and only their example answers: as ex-2.2.22 found, nearly every query answer has a posterior near 1, so there is too little spread to compare there. E4 also uses the held-out sets with 1 to 5 examples, for the task score at the query.

The `k-mixed` run at model seed 702 has no anchor at the example answers: α stays near the control even where the example fits `difference`. It also scores lowest on the task of all the runs here, as the runs that took the slow path through training did in ex-2.2.21. It is left out of the numbers below.

The measurements

α is the alignment with e_1 (the cosine between a state and the anchored direction), averaged over slices 2 to 4, at the answer of each example. Slice 1 shows the same pattern, more weakly, and the embedding shows none.

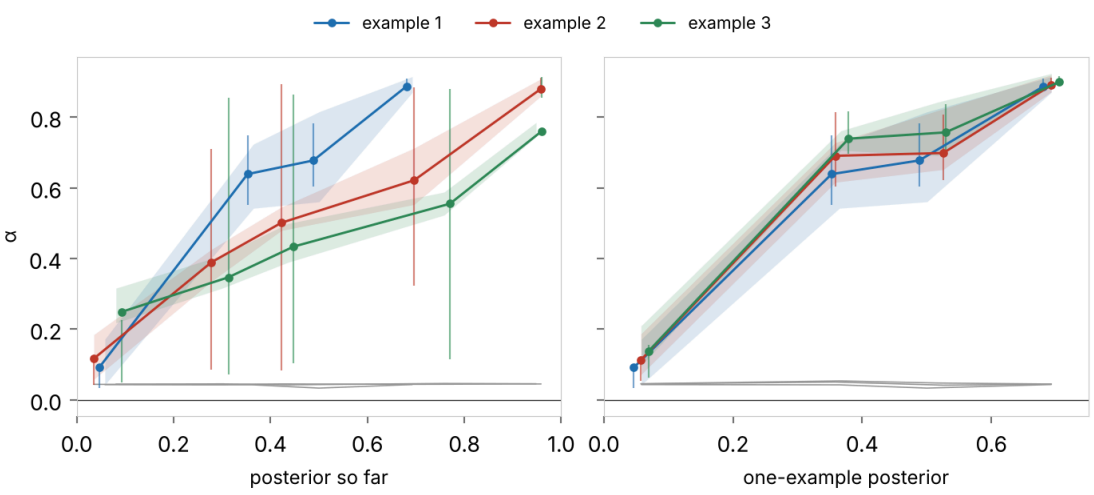
Each example answer is compared with three summaries of the evidence, all on the noise model of the corpus:

- the posterior on `difference` given the examples up to and including that one (*the posterior so far*, the one ex-2.2.22 used);
- the posterior given that example alone, from a uniform prior over the seven ops (*the one-example posterior*);
- the posterior given only the examples before it (*the earlier posterior*), which is 1/7 at the first example.

About two thirds of the examples in a `difference` context *fit* it, meaning the answer shown is one `difference` can give on that pair. The rest come from replacement op noise or a random color. A single example can lift the posterior on `difference` from 1/7 to about 0.7 at most, since other ops can give the same answer on some pairs. An example that does not fit has a one-example posterior near zero.

Each example on its own (E1)

If α followed the posterior so far, answers with the same posterior would have the same α whatever their position. The left panel below repeats the comparison of ex-2.2.22 on `hinge`, and the right panel plots the same answers against the one-example posterior.



α against two summaries of the evidence, on `hinge`. *At each example answer of the held-out `difference` contexts, α averaged over slices 2 to 4, against* ***left:*** *the posterior so far, and* ***right:*** *the one-example posterior, in bins of 0.2, each point at the mean evidence of its bin. One line per answer index, the seed mean of six runs; a bin with fewer than 20 points in a run is left out of that run. The band is the seed range of the run means (pooled over contexts first); the whiskers span the middle half of the contexts, on α averaged over the runs (pooled over seeds first). Thin grey lines: the control.*

On the left the three lines are offset, as in ex-2.2.22: at a posterior between 0.6 and 0.8, α at the first answer is about as high as it gets, and at the third it is about a third lower. On the right they lie on top of each other. The seed bands are narrow on both sides. The whiskers are not: at the same posterior so far, α spreads over most of its range from one context to the next, and against the one-example posterior that spread shrinks to a few tenths or less. An answer whose example fits `difference` has α near 0.88 at every index, and one that does not fit stays low.

That explains the offset. At the first answer, a middling posterior means one example that fits `difference` and also fits another op. At the third, it usually means a mix of examples that fit and examples that do not, and α then depends on which kind the latest one is.

The table gives the same comparison as variance explained, for each anchored condition. The share of the examples so far that fit, a summary that weights every example equally, does a little better than the posterior so far and much worse than the one-example posterior.

condition	runs	posterior so far	share that fit	one-example posterior
`hinge`	6	0.47 (0.46–0.48)	0.57 (0.55–0.60)	0.92 (0.87–0.96)
`whole-line`	3	0.46 (0.44–0.47)	0.54 (0.51–0.57)	0.89 (0.85–0.91)
`k-mixed`	2	0.48 (0.47–0.48)	0.59 (0.57–0.60)	0.92 (0.89–0.96)

How much of the variance in α each summary accounts for. *The squared correlation r^2 between α at the example answers (averaged over slices 2 to 4) and each summary of the evidence, over every example answer of the held-out `difference` contexts. Seed mean, with the seed range in brackets.*

What the earlier examples add (E2)

The one-example posterior leaves some variance unexplained, so the earlier examples might still add something. A least-squares fit of α on the one-example posterior and the earlier posterior together gives each a weight; both are on the same scale, from 0 to 1.

condition	weight, this example	weight, earlier examples	α, misfit after doubt	α, misfit after support
`hinge`	1.22 (1.13–1.26)	0.04 (0.02–0.06)	0.09 (0.05–0.17)	0.16 (0.10–0.22)
`whole-line`	1.33 (1.29–1.36)	0.05 (0.05–0.06)	0.06 (0.05–0.07)	0.14 (0.13–0.16)
`k-mixed`	1.22 (1.21–1.23)	0.04 (0.02–0.05)	0.09 (0.08–0.10)	0.15 (0.14–0.16)

The weight of each example and of the ones before it. *The first two columns: the least-squares weights of α on the one-example posterior and on the earlier posterior. The last two: mean α at answers whose example does not fit* `difference`*, when the earlier posterior is below 0.3 and when it is above 0.8. Seed mean, with the seed range in brackets.*

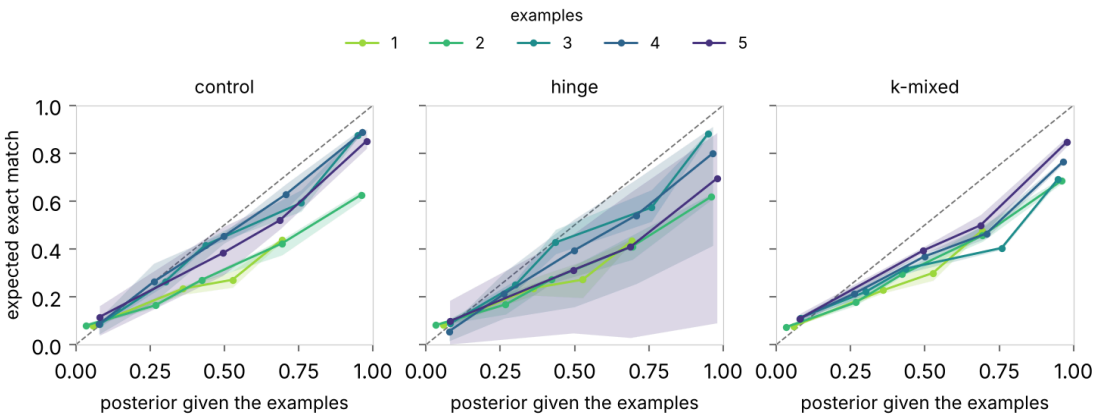
The earlier examples get a weight of a few hundredths, against more than one for the example itself. They show most on the answers that do not fit: there α is a little higher when the earlier examples favor `difference` than when they do not (0.16 against 0.09 on `hinge`). On answers that fit, the earlier examples make almost no difference.

The answer color alone (E3)

The one-example posterior depends on the operands as well as the answer. If α could be predicted from the answer color alone, the anchor would be closer to a property of the answer token. Averaged by answer color on half of the contexts and tested on the other half, the answer color accounts for 0.09 (0.06–0.14) of the variance on `hinge` (held-out R^2), and similar on the other conditions. So the model places the state at an example answer on e_1 according to how that answer relates to its operands.

Predicting the next answer (E4)

The anchor follows each example on its own, but the next-token loss asks the model to predict each example answer at its `=`, which means combining the examples before it. How well does it do that? The by-count pass scored the query on held-out sets with 1 to 5 examples. The query after one example asks for the answer a second example would show (without the noise an example can carry), and the query after two asks for the third. So these scores stand in for how well the model predicts the answers of the later examples. The control and `hinge` were trained on three examples only, so for them the other counts are outside training; `k-mixed` was trained on 1 to 5.



How well the model predicts the next answer. *On the held-out `difference` contexts with 1 to 5 examples, expected exact match at the query against the posterior on `difference` given the examples, in bins of 0.2, each point at the mean posterior of its bin. One line per number of examples; the seed mean, with the seed range as a band. Dashed: the ideal predictor, which gets the posterior right. The control and `hinge` were trained on three examples only.*

Expected exact match. The task score, as the chance that a drawn answer agrees with a true one....

The ideal predictor gets an expected exact match about equal to its posterior, so it follows the diagonal. With three examples, the count they were trained on, the control and `hinge` come close to it: 0.88 against 0.95 in the top bin on `hinge`. With one example every condition falls well short, at about 0.43 against 0.70, and `k-mixed`, which was trained on that count, does only a little better (0.48). With two examples the gap is wider still: 0.62 against 0.96 on `hinge`, and 0.68 on `k-mixed`. `k-mixed` stays below the diagonal at every count, reaching 0.69 with three examples. A single example lifts the posterior to about 0.7 at most, and with three examples the control and `hinge` score well above that, so at the query the model does combine the examples. The anchor at the example answers does not follow that combining.

Discussion

At the example answers the anchor marks examples that look like `difference` more than contexts that do. The rise with the posterior that ex-2.2.22 saw comes mostly from this: the more examples fit, the higher the posterior, and each fitting example has a high α of its own. So this measurement does not yet show the anchor holding an inferred op. The sites where that could show are the `=` tokens, where the model predicts an answer it has not yet seen. The next-token loss covers every one of them, so the model is asked to combine the examples so far at each example `=` as well as at the query `=`. The answer, one token later, is where the example just shown is in view, which is what α follows. Ex-2.2.21 found the anchor weak at the query `=`, and α at the example `=` tokens was not stored.

This fits how the label works. The pull is pooled over the positions of a whole `difference` context, so the model can meet it at positions of its choosing, and the answers of the examples that fit are where a single line shows the op most plainly. The answers that do not fit stay low, which a pooled pull allows.

It may also bear on the spill that ex-2.2.23 found. A per-example judgement of whether an answer fits `difference` given its operands would fire on examples of other ops whose answers happen to fit, and removing e_1 there would touch those ops. This re-analysis covers only `difference` contexts, so it does not test that.